

Comparative analysis of heart failure detection via machine learning

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Abstract—Neural networks are often considered as the “go to” machine learning model due to their usually accurate predictions. However, can this model perform more accurately compared to simpler, classical models on important real-world data? This project aims to answer this question by comparing neural networks against classical models to predict heart failure in patients.

Keywords— *Heart failure; neural networks; decision trees; random forest; gradient-boosting; logistic regression; evaluate;*

I. INTRODUCTION

Cardiovascular diseases are the leading cause of death worldwide [1], contributing to ~681,000 deaths annually in the United States alone. Heart failure is a common condition caused by cardiovascular diseases, and early detection is necessary for preventing deaths from this health issue.

Traditionally, clinicians evaluate the risk of cardiovascular disease using methods like Logistic Regression models or Framingham Risk Scores [2]. These methods, while interpretable, often struggle to digitize complex, non-linear interactions found in clinical tabular data, such as correlations between physiological features and lifestyle factors. Random Forest decision tree models and Neural Networks have thus been adopted in the field, aiming to improve prognostic accuracy by better defining these complex correlations.

II. OBJECTIVE

The goal of this project is to perform a comparative analysis of classical machine learning models versus modern neural network-based approaches in predicting heart failure outcomes using clinical data. We will aim to evaluate the performances of models such as decision trees, random forest, logistic regression, gradient boosting, and neural networks using our dataset.

III. APPROACH

Our approach can be split into the following four sections:

A. Data Preprocessing

Once retrieved, the dataset will be explored to get a better sense of the data being represented to determine if any data is missing. If needed, features will be scaled and normalized, after which the data will be split into training and testing sets

B. Model Implementation

The project will implement both classical and modern, state-of-the-art models. The classical models will include logistic regression, decision trees such as random forest and gradient boosting. The state-of-the-art model will be a neural network (multi-layer perceptron) implementation.

C. Model Evaluation

Each model's performance will be evaluated using accuracy, precision, recall, F1-scores, ROC-AUC scores, and confusion matrixes. Additionally, each model's tendency to overfit, generalize, and training time will be recorded.

D. Analysis and Comparison

The performances of each classical model will be compared with one another and to neural networks. These comparisons will determine which model performed the best and whether or not neural networks provide the significant advantage that they have shown. These comparisons could also highlight patterns within the actual datasets and determine which clinical features are significant predictors for heart failure.

IV. DATASET

This project's dataset is titled "Heart Failure Clinical Records" and was found on the UC Irvine Machine Learning Repository website at

<https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records>.

The dataset contains medical records of 299 patients who experienced heart failure and was collected during follow-up appointments. 13 clinical features including blood-related measurements are included to describe each patient's medical condition.

V. TIMELINE

(Dates are subject to change)

TABLE I. TIMELINE

Date	Task
February 13, 2026	Finish and submit project proposal
February 14, 2026	Begin data preprocessing
February 28, 2026	Finish data preprocessing; begin developing machine learning models
February 28, 2026	Make project progress presentation
March 2, 2026	Present project progress presentation
April 6, 2026	Finish developing models; begin evaluating
April 20, 2026	Begin final project report
April 27, 2026	Present final presentation; submit final report

Fig. 1. Project tasks and due dates

VI. ROLES

A. Data Engineer

The Data Engineering lead is responsible for the foundational data pipeline, ensuring the integrity and quality of the dataset prior to model ingestion. This role conducts comprehensive exploratory data analysis to identify underlying distributions, feature correlations, and potential data leakage risks, particularly concerning the time variable.

B. Classical Models Developer

The classical modeling lead is tasked with developing a robust traditional machine learning model, to serve as a high-performing comparative baseline against the neural network. This role involves engineering the algorithm to optimally handle the tabular clinical data and executing hyperparameter optimization through techniques to determine the most optimal way to structure the model.

C. Neural Network Developer

The Neural Network Developer lead oversees the design, implementation, and optimization of the Artificial Neural Network. This role focuses on constructing a Multi-Layer Perceptron that effectively captures complex, non-linear patient patterns, as well as finding the optimal number of hidden layers and neuronal density.

D. Evaluator

The Evaluator moves beyond baseline accuracy to evaluate the models using advanced classification metrics. This role interprets false-positive and false-negative rates within a real world clinical context. They also generate comparative visualizations.

REFERENCES

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